

Parametrization of Trivalent and Tetravalent Metal Ions for the OPC3, OPC, TIP3P-FB, and TIP4P-FB Water Models

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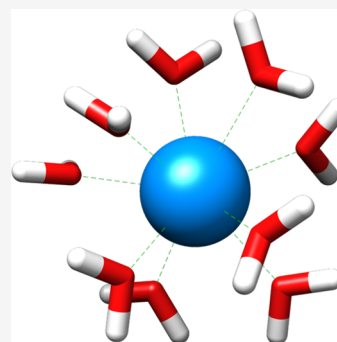
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ABSTRACT: Commonly seen in rare-earth chemistry and materials science, highly charged metal ions play key roles in many chemical processes. Computer simulations have become an important tool for scientific research nowadays. Meaningful simulations require reliable parameters. In the present work, we parametrized 18 M(III) and 6 M(IV) metal ions for four new water models (OPC3, OPC, TIP3P-FB, TIP4P-FB) in conjunction with each of the 12-6 and 12-6-4 nonbonded models. Similar to what was observed previously, issues with the 12-6 model can be fixed by using the 12-6-4 model. Moreover, the four new water models showed comparable performance or considerable improvement over the previous water models (TIP3P, SPC/E, and TIP4P_{EW}) in the same category (3-point or 4-point water models, respectively). Finally, we reported a study of a metalloprotein system demonstrating the capability of the 12-6-4 model to model metalloproteins. The reported parameters will facilitate accurate simulations of highly charged metal ions in aqueous solution.



+3 & +4 Metal Ions

12-6-4 Interactions

New Water Models:

OPC3
OPC
TIP3P-FB
TIP4P-FB

INTRODUCTION

Highly charged (trivalent and tetravalent) metal ions are of great interest in both rare-earth chemistry¹ and materials science.² Due to a high charge-to-size ratio, trivalent and tetravalent metal ions usually have special properties as coordination centers, which have been utilized in optical engineering and enzymatic catalytic processes.^{3–5} For instance, only one single charge difference between Fe(II) and Fe(III) in a heme molecule will cause a dramatic change in its affinity toward FAD, a flavin-derived oxidoreductase cofactor. This fine-tuning between metal ion charge and biological activity is crucial for rescuing and maintaining normal cell function after damage from ultraviolet (UV) light or free radicals.⁶ Along with the increasing market for modern electronical devices, research on lanthanide series metals has been thriving for years. Due to the unique structure and magnetic spin states of 4f electrons, lanthanide series metals, such as Nd and Y, have been used in many material complexes that function as luminescent gels or laser crystals.⁷ Meanwhile, the actinide series elements, like Th and U, are well known for their radioactivity,⁸ but their chemical properties as +4 metal ions are also important to guarantee safety during excavation, utilization, and disposal. For example, due to its uniquely low vapor pressure and high chemical stability, uranium tetrafluoride (UF₄) is widely used as a key component in molten salt nuclear reactors.⁹ In short, many elements can form highly charged metal ions, but physical chemists face the major challenge to accurately model them computationally.

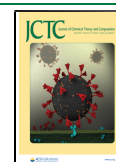
Several theoretical methods to model metal ions have been well developed and established over the past decades: the

quantum mechanics (QM) method,^{10,11} the molecular mechanics (MM) method,^{12–26} and the QM/MM hybrid method.^{27,28} In view of its computational speed, classical force fields have a considerable advantage over QM-based methods,^{29,30} because instead of explicitly treating electrons, classical force fields use simple analytical functions to model the molecular interactions. Although concerns still remain on the precision of MM methods to study chemical systems at the molecular level,³¹ these concerns can be mitigated via the development of realistic interactions and molecular models, which can then have their transferability validated by simulating realistic chemical and biological systems.

For models describing interactions between metal ions and their surrounding molecules, the bonded model,^{17,32,33} nonbonded model,³⁴ and cationic dummy atom model³⁵ are three widely used ones. The bonded model represents the interactions between the ion and its surrounding residues via terms describing the bond, angle, torsion, Coulombic, and van der Waals (VDW) interactions. Since the harmonic approximation is used in the bonded model, many processes that form or break chemical bonds cannot be simulated.¹⁷ In contrast, the

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nonbonded model only assigns a charge on the metal ion of interest and then uses Coulombic and VDW terms to represent the interactions surrounding the metal ion. Since the coordinate bond is simplified as merely two types of interactions, the nonbonded model tends to underestimate the interaction energy significantly, especially when the ion size is small or the charge/size ratio of the metal ion is high.³⁴ As for the cationic dummy atom model, it combines the bonded and nonbonded models by placing arbitrarily defined charge points around the metal ion through covalent bonds, guiding the surrounding molecules to coordinate according to the geometry of the added charge points.²³ This, however, may face difficulties when the metal ion changes its coordination number in different surroundings (e.g., from water to a protein binding site).³⁵ Apart from these three models, polarizable force fields have also been developed to simulate metal ions, but it requires careful parametrization to define the polarization effect between the ion and its surroundings.^{36–42}

Although it is not the most accurate model to describe metal ion interactions, the 12-6 Lennard-Jones (LJ) nonbonded model is still widely used because of its simplicity, computational efficiency, and transferability between various molecular environments.^{14,16,43} Moreover, a charge scaling model has been described based on the traditional 12-6 Lennard-Jones potential and had shown promising results in metal ion simulations.^{44–48} Meanwhile, in earlier research, we found that due to the neglect of the ion-induced dipole interactions, the LJ nonbonded model cannot reproduce hydration free energy (HFE) and ion–oxygen distance (IOD) for a certain ion simultaneously when the ion charge is $\geq +2$.⁴⁹ For example, if the IOD is reproduced, the HFE can be tens of kcal/mol less negative than the experimental value for a divalent metal ion.³⁴ Alternatively, reproducing HFEs will yield IOD values that are about 0.27 Å shorter than experimental values for divalent ions on average.³⁴ To solve this issue, we proposed a 12-6-4 LJ-type nonbonded model to account for the ion-induced dipole interaction for metal ions.⁵⁰ By adding a C_4 term coupled with suitable optimization, this model can reproduce the HFE and IOD of a metal ion simultaneously.^{50,51} Moreover, further applications have demonstrated that the 12-6-4 LJ-type nonbonded model can be applied to more complex cases, like nucleic acid systems and metalloproteins in a salt solution.^{52,53}

Given the capability of the 12-6-4 model to represent the interactions between metal ions and their environment, previously we parametrized it for 24 highly charged metal ions (18 M(III) ions and 6 M(IV) ions)⁵⁴ in conjunction with three water models (TIP3P,⁵⁵ SPC/E,⁵⁶ and TIP4P_{EW}⁵⁷). Our results illustrated a much larger underestimation of ion–water interactions for the highly charged ions while using the 12-6 model than for the monovalent and divalent metal ions, which is likely due to the stronger dipole-inducing ability of the higher charged ions.^{34,49,54} In contrast, the 12-6-4 model could simultaneously reproduce the experimental HFE and IOD values for all of the highly charged metal ions.⁵⁴

In the present work, we parametrized the same 24 highly charged metal ions for four new water models (OPC3,⁵⁸ OPC,⁵⁹ TIP3P-FB,⁶⁰ and TIP4P-FB⁶⁰). These four new water models have shown advantages for simulating biological systems in an explicit water environment. Specifically, the OPC3 and OPC water models were designed to reproduce the bulk properties of water. By using either of them, the HFEs of small molecules can be reproduced within ± 1 kcal/mol of the experimental values.⁶¹ In contrast, the TIP3P-FB and TIP4P-FB water models can

accurately reproduce the liquid bulk properties over a wide range of temperatures due to the inclusion of force balance consideration when parametrizing these models.⁶⁰ Since ion parameters are water model dependent, and no parameters have been reported for simulating highly charged metal ions in conjunction with these four new water models, we decided to parameterize 12-6 and 12-6-4 models for highly charged metal ions for these water models in conjunction with the particle mesh Ewald (PME) method. The obtained parameters reproduce experimental HFE and IOD values and showed reasonable agreement with quantum mechanical (QM) data.^{62,63}

METHODS

Potential Functions. In the 12-6 nonbonded model, the Coulombic interactions are considered along with the 12-6 LJ potential leading to the formula shown in eq 1.

$$U_{ij}(r_{ij}) = \frac{e^2 Q_i Q_j}{r_{ij}} + \epsilon_{ij} \left[\left(\frac{R_{\min,ij}}{r_{ij}} \right)^{12} - 2 \left(\frac{R_{\min,ij}}{r_{ij}} \right)^6 \right] \quad (1)$$

where U_{ij} is the potential of interatomic interactions between atoms i and j as a function of the interatomic distance r_{ij} , e is the charge of a proton, Q_i and Q_j are the charges of the metal ion and an interacting atom, respectively. Since the metal ion and water molecules all have fixed charges (the metal ion has an integer charge consistent with its oxidation state, while the water model has its partial charges fixed as well), the only two parameters being tuned in this 12-6 nonbonded model are $R_{\min,ij}$ and ϵ_{ij} . In AMBER, the Lorentz–Berthelot combining rules are used to obtain the $R_{\min,ij}$ and ϵ_{ij} . The details of these combining rules are given in eqs 2 and 3. Herein, $R_{\min,ii}/2$ and $R_{\min,jj}/2$ correspond to the VDW radii of atoms i and j , and represent the VDW potential well depths of atoms i and j , respectively.

$$R_{\min,ij} = \frac{R_{\min,ii} + R_{\min,jj}}{2} \quad (2)$$

$$\epsilon_{ij} = \sqrt{\epsilon_i \times \epsilon_j} \quad (3)$$

For metal ions, we treated the $R_{\min}/2$ and ϵ as two correlated variables connected by the noble gas curve (NGC)³⁴

$$\epsilon = 10^{-57.36 \times e^{-2.471 \times R_{\min}/2}} \quad (4)$$

The partial charges of different water models, together with their $R_{\min}/2$ and ϵ values for the water oxygen atom, are listed in Table S1.

In comparison to the 12-6 nonbonded model, the 12-6-4 LJ-type nonbonded model has an additional C_4 term, which requires one additional variable C_4

$$U_{ij}(r_{ij}) = \frac{e^2 Q_i Q_j}{r_{ij}} - \frac{C_4}{r_{ij}^4} + \epsilon_{ij} \left[\left(\frac{R_{\min,ij}}{r_{ij}} \right)^{12} - 2 \left(\frac{R_{\min,ij}}{r_{ij}} \right)^6 \right] \quad (5)$$

By definition of the interaction between a point charge and an induced dipole,⁶⁴ the expression of C_4 is given by

$$C_4 = \frac{\alpha(z\epsilon)^2 \cos \theta}{2(4\pi\epsilon_0\epsilon)^2} \quad (6)$$

where α is the polarizability of the ligating atom. z is the charge of the metal ion. θ is the angle between the charge-induced

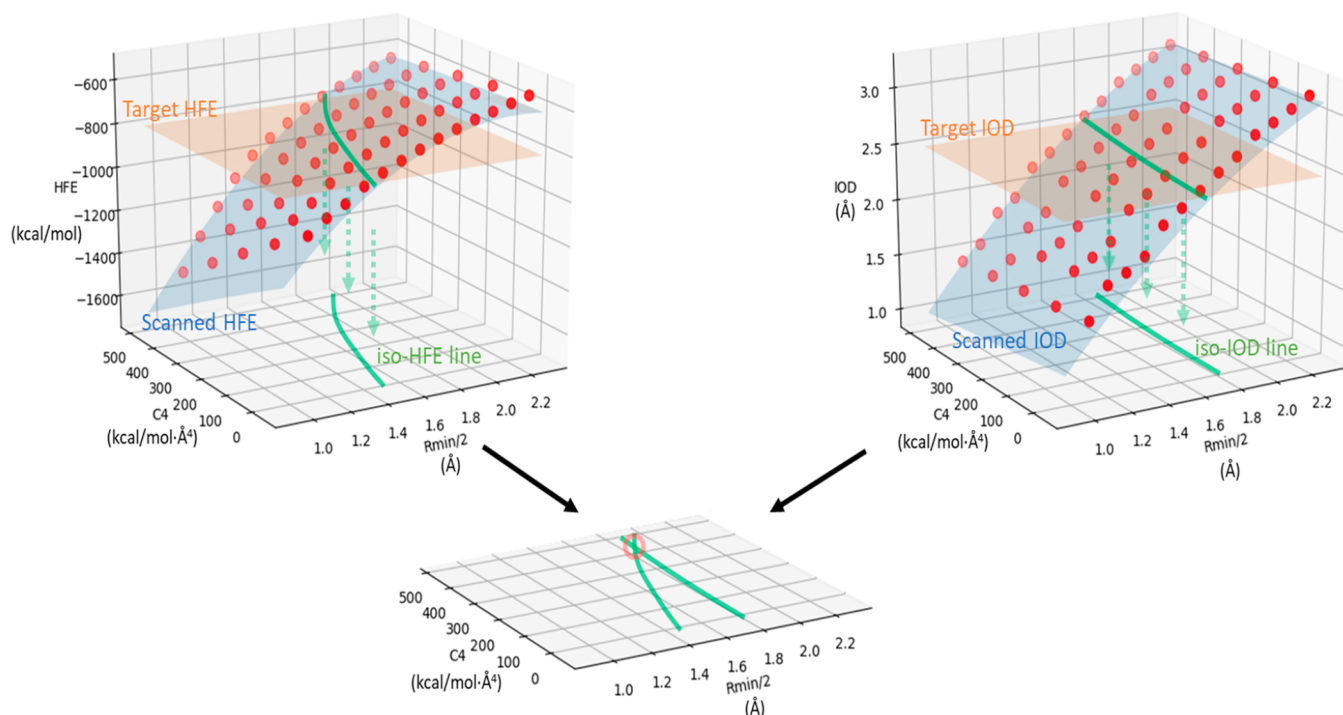


Figure 1. Illustration of the scan-and-determine parametrization strategy (see Tables S2 and S3 for the final results).

dipole center line and direction of the induced dipole, which is usually close to π . $\epsilon_0\epsilon$ gives the electron permittivity between the point charge and induced dipole. The combining rule of C_4 has been discussed in our previous work.⁴⁹

Comparing eq 5 to eq 1, the added C_4 term will lower the potential energy between two interacting atoms if the electrostatic and VDW terms do not change. This is consistent with the physical meaning of the C_4 term which corresponds to the ion-induced dipole interactions.

In previous parametrization efforts for TIP3P, SPC/E, and TIP4P_{EW}, we observed a significant water model dependency of the C_4 term.^{49,50,54} This implies that for the OPC3, OPC, TIP3P-FB, and TIP4P-FB water models, a reparametrization of the C_4 term is necessary. Hence, for the 12-6-4 parametrization of a metal ion for a specific water model, we have two variables $R_{\min}/2$ and C_4 that need to be determined. In the present work, we treat HFE and IOD as the two target properties, which can be fitted to functions (see eq 7) allowing us to estimate the parameters based on the parameter scan results.

$$\begin{cases} \text{HFE} = f(R_{\min}, C_4) \\ \text{IOD} = g(R_{\min}, C_4) \end{cases} \quad (7)$$

This is illustrated in Figure 1 where the red dots are the scanned values of HFE and IOD as a result of varying the $R_{\min}/2$ and C_4 inputs. The curved blue surfaces were fitted based on the scan results with $R_{\min}/2$ in the range of 1.1–2.3 Å, and the flat brown planes are the target values for HFE and IOD. The resultant projections of the intersecting curves between the blue surfaces and brown planes (i.e., the projected green curves) crossover at the orange circle, yielding our initial guesses for $R_{\min}/2$ and C_4 parameters for various metal ions. Afterward, the parameters were fine-tuned to meet the desired accuracy.

Calculation of HFE, IOD, and CN values. To calculate the HFE, the thermodynamic cycle of Figure 2 was used.

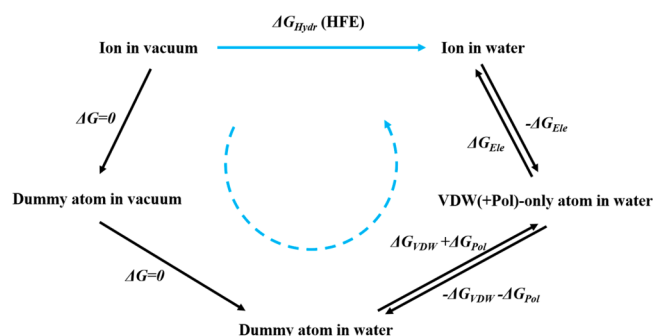


Figure 2. Thermodynamic cycle for simulating the solvation process of a metal ion.

In Figure 2, the blue solid arrow represents the desired free energy, and the blue dashed arrow indicates the indirect pathway used to obtain the HFE. The bidirectional black arrows indicate that the free energies were calculated twice, both forward and backward to better estimate the free energy values. The calculations of $\Delta G_{\text{VDW}} + \Delta G_{\text{Pol}}$ and ΔG_{Ele} were accomplished using thermodynamic integration (TI),⁶⁵ where ΔG_{Pol} is for the ion-induced dipole interaction which is represented by the C_4 term. Herein, we employed the same protocol for simulating HFE as we did in our research for the parametrization of divalent metal ions.⁵¹ The IOD and CN values were obtained based on the radial distribution function (RDF), as we described in our previous study.⁵⁰

Calculation of the Diffusion Coefficients of Several Metal Ions. To further evaluate the performance of our parameters, the diffusion coefficients of Fe^{3+} , Al^{3+} and Cr^{3+} were calculated for the optimized 12-6 HFE, 12-6 IOD, and 12-6-4 parameter sets with the OPC water model. Using a protocol adapted from Panteva et al.⁶⁶ and Izadi et al.,⁵⁹ the system was simulated at different concentrations by changing the water box sizes. Specifically, a metal ion was first solvated by four quasi-

cubic boxes with box lengths of 31, 40, 50, and 62 Å, respectively. Then, we employed the same protocol as in our previous study,⁵¹ with the details described in the Supporting Information of the previous study.

RESULTS AND DISCUSSIONS

Experimental HFE and IOD Values of Trivalent and Tetravalent Ions. Extending our previous work,⁵⁴ we parametrized 24 highly charged metal ions in conjunction with four new water models. The target HFE and IOD values of these 18 M(III) and 6 M(IV) ions are listed in Table 1. The HFE values

Table 1. Experimental HFE, IOD, and CN of the First Solvation Shell for 18 Trivalent and 6 Tetravalent Ions^a

Ions	HFE (kcal/mol)	IOD (Å)	CN	effective ion radii (Å)	first shell water radii (Å)
Al ³⁺	−1081.5	1.88	6	0.54	1.34
Fe ³⁺	−1019.4	2.03	6	0.65	1.38
Cr ³⁺	−958.4	1.96	6	0.62	1.34
In ³⁺	−951.2	2.15	6	0.80	1.35
Tl ³⁺	−948.9	2.23	4–6	0.89	1.34
Y ³⁺	−824.6	2.36	8	0.90	1.46
La ³⁺	−751.7	2.52	8.0–9.1	1.03	1.49
Ce ³⁺	−764.8	2.55	7.5	1.01	1.54
Pr ³⁺	−775.6	2.54	9.2	0.99	1.55
Nd ³⁺	−783.9	2.47	8.0–8.9	0.98	1.49
Sm ³⁺	−794.7	2.44	8.0–9.9	0.96	1.48
Eu ³⁺	−803.1	2.45	8.3	0.95	1.50
Gd ³⁺	−806.6	2.39	8.0–9.9	0.94	1.45
Tb ³⁺	−812.6	2.40	8.0–8.2	0.92	1.48
Dy ³⁺	−818.6	2.37	7.4–7.9	0.91	1.46
Er ³⁺	−835.3	2.36	6.3–8.2	0.89	1.47
Tm ³⁺	−840.1	2.36	8.1	0.88	1.48
Lu ³⁺	−840.1	2.34	8	0.86	1.48
Hf ⁴⁺	−1664.7	2.16	8	0.85	1.31
Zr ⁴⁺	−1622.8	2.19	8	0.86	1.33
Ce ⁴⁺	−1462.7	2.42	9	0.87	1.55
U ⁴⁺	−1567.9	2.42	9–11	0.89	1.53
Pu ⁴⁺	−1520.1	2.39	8	0.86	1.53
Th ⁴⁺	−1389.8	2.45	9–11	0.84	1.51

^aThese HFE, IOD, and CN values are from ref 54, which cites the original work.

from Marcus⁶⁷ are used here in order to be consistent with our previous work,^{51,68} and as we have discussed that the Marcus data set represents the “intrinsic HFE”, which is consistent with our simulations using periodic boundary conditions (PBCs).^{51,68} However, if one wants to use a different HFE target value, one can perform the parametrization based on the parameter scan results presented in the SI.

To briefly summarize Table 1, the target HFE and IOD values showed a monotonic trend inside each of the IIIA group metal ions (Al³⁺, Tl³⁺, and In³⁺), the IIIB group metal ions (Y³⁺ and La³⁺), and the IVB group metal ions (Zr⁴⁺ and Hf⁴⁺). In terms of Cr³⁺ and Fe³⁺, which are transition metal ions in the same period, their different electronic configurations give them different HFEs and IODs. Besides these, all other metal ions belong to the lanthanide and actinide series, which have similar, yet subtly different, HFE and IOD values. These differences are further discussed below with the final 12-6-4 parametrization results.

Parameter Scanning Results for 12-6 and 12-6-4 Models. The scanning was carried out in a pattern described in Figure 1. Since both trivalent and tetravalent metal ions have a similar range of sizes, the scanning $R_{\min}/2$ ranges were both 0.9–2.3 Å with an interval of 0.1 Å. However, due to the differences in charges, the ability to induce a dipole will be significantly different between trivalent and tetravalent metal ions, as indicated by eq 6. Therefore, in terms of parameter scanning, C_4 values were set to a range of 0–500 kcal/mol·Å⁴ with an interval of 125 kcal/mol·Å⁴ for the trivalent ions, while a 0–1000 kcal/mol·Å⁴ range with an interval of 250 kcal/mol·Å⁴ was used for the tetravalent ions. All scanning results are presented in Tables S2 and S3.

The scanning was conducted on three input variables (i.e., water model, $R_{\min}/2$, and C_4), but to simplify the analysis, the influence of the water model and $R_{\min}/2$ value on the scan results (with $C_4 = 0$) were first analyzed in Figure 3.

In order to get a better sense of differences between the three previous water models (TIP3P, SPC/E, and TIP4_{EW}) and the four new water models (OPC3, OPC, TIP3P-FB, and TIP4P-FB), the scanning results of all seven water models are plotted together in Figure 3, with the scanning results for the three previous water models taken from ref 54. By comparing within Figure 3A or C, one can conclude that when the $R_{\min}/2$ values are the same, OPC3 usually gives the lowest HFE for both trivalent and tetravalent metal ions, especially when the $R_{\min}/2$ values are close to 1.0 Å. Generally, the three-point water models (TIP3P, SPC/E, OPC3, and TIP3P-FB) give more negative HFEs than four-point water models (TIP4P/EW, OPC, TIP4P-FB) while using the 12-6 model only (i.e., $C_4 = 0$). This phenomenon matches what was observed previously.⁵¹

However, when comparing within Figure 3B or D, the IOD differences between water models is not as significant as the HFE differences, but the three-point water models still have a systematically lower IOD when compared to four-point water models at the same $R_{\min}/2$ value, especially when the $R_{\min}/2$ value is close to 2.3 Å.

Next, the effect of C_4 term was further analyzed in Figure 4 by plotting IOD against HFE for three data groups: (1) the targeted HFE and IOD of the metal ions, which are represented by black dots, (2) the scanned HFE-IOD relationship when $C_4 = 0$, which are represented by solid curves, and (3) the scanned HFE-IOD relationship with C_4 taking its maximum in the scanned range (i.e., 500 kcal/mol·Å⁴ for trivalent metal ions and 1000 kcal/mol·Å⁴ for tetravalent metal ions), which are represented by long dashed curves.

Still, all three-point water models showed a systematic difference compared to four-point water models. In short, at the same HFE, the three-point water model will give larger IOD values, or at the same IOD, the three-point water model will give a lower HFE value. This is consistent with Figure 3, where three-point water models always have lower HFE and lower IOD at the same $R_{\min}/2$ value compared to four-point water models. As an outcome, the OPC3 water model has the best performance when the system only uses the 12-6 model, since the HFE vs IOD curve of OPC3 when $C_4 = 0$ is the closest to the target experimental values for both trivalent and tetravalent metal ions. In general, the three-point water models showed better performance than the four-point water models. Moreover, the four new water models have comparable or significantly better performance than the previous water models in the same category (by making a comparison inside the three-point water models and inside the four-point water models, respectively).

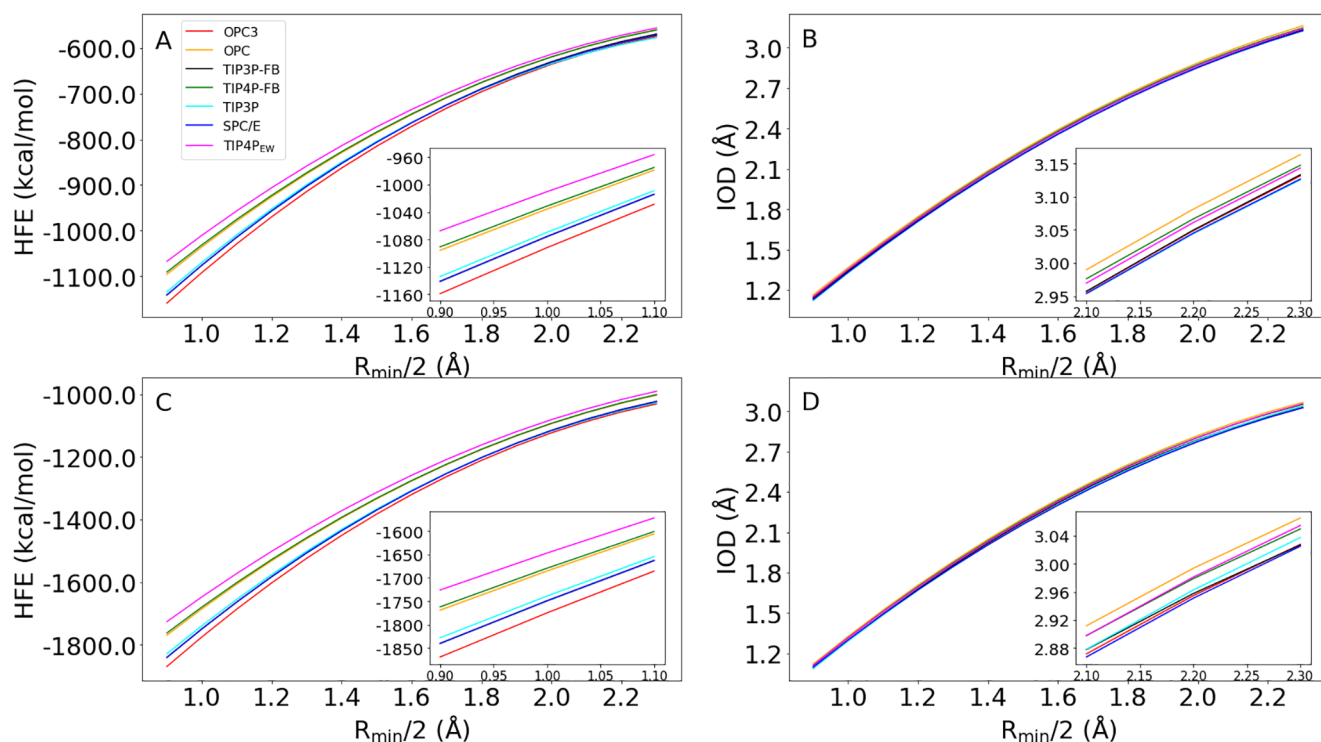


Figure 3. 12-6 HFE fitting curves for the seven water models for trivalent (A) and tetravalent (C) ions and the 12-6 IOD fitting curves for the seven water models for trivalent (B) and tetravalent (D) ions. The fitting curves for the TIP3P-FB and SPC/E water models overlap with each other in panels A and C.

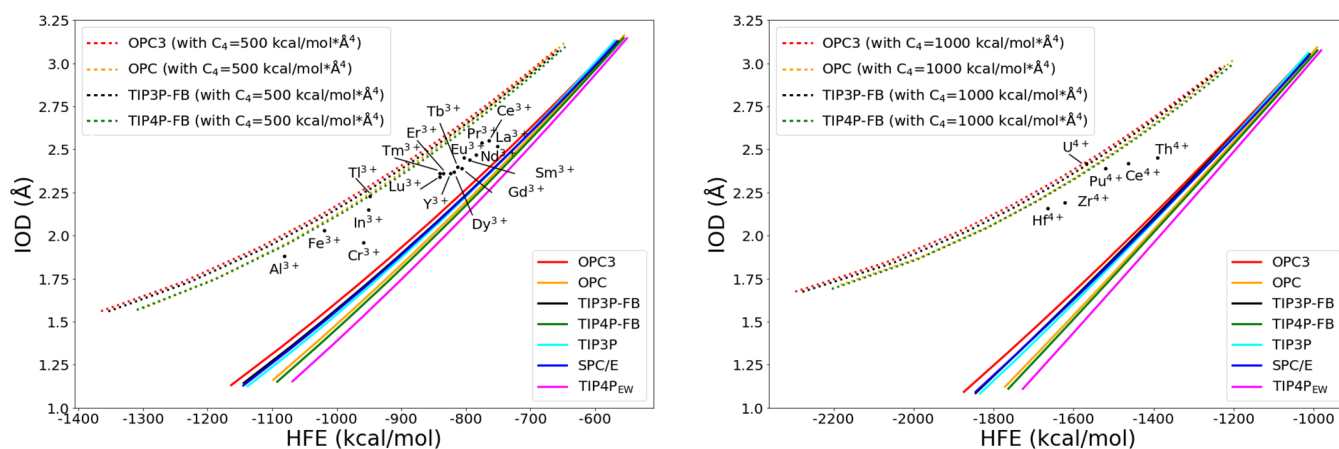


Figure 4. IOD vs HFE scan results of each water model and their comparisons with experimental IOD vs HFE for the trivalent metal ions (left) and tetravalent ions (right).

These observations agree with our previous study of divalent metal ions.⁵¹

Also, from each subplot in Figure 4, one observes that for both trivalent and tetravalent metal ions, the target experimental values are “sandwiched” by the two sets of curves representing the minimum and maximum of the C_4 value in the scan range. This indicates that through using a suitable C_4 parameter, the 12-6-4 model is able to reproduce both experimental HFE and IOD values of a highly charged metal ion simultaneously. Starting from the initial guesses obtained by the strategy described in Figure 1, we performed the final parameter determination of the 12-6-4 model for all 24 metal ions.

Parameterization of the 12-6 Model. The parameterization for the 12-6 model (with $C_4 = 0$) was first carried out for all

the 24 metal ions. Since the 12-6 model cannot reproduce both HFE and IOD at the same time regardless of the $R_{\min}/2$ value chosen for the trivalent and tetravalent metal ions, the parametrization was performed in two parts: determination of two parameter sets that reproduce HFES and IODs, respectively. The two parameter sets are labeled as the 12-6 HFE and 12-6 IOD sets. The 12-6 HFE and 12-6 IOD parameter sets are initially guessed based on Figure 3. They were then fine-tuned to reach desirable accuracy: within 2 kcal/mol of experimental HFES for the 12-6 HFE parameter set and within 0.01 Å of the experimental IODs for the 12-6 IOD parameter set. The optimized 12-6 HFE and 12-6 IOD parameter sets are listed in Tables 2 and 3, respectively. From Tables 2 and 3, it is clear that the HFE and IOD parameters sets are intrinsically different,

Table 2. 12-6 HFE Parameter Set for 18 Trivalent and 6 Tetravalent Metal Ions in Conjunction with OPC3, OPC, TIP3P-FB, and TIP4P-FB Water Models

	OPC3		OPC		TIP3P-FB		TIP4P-FB	
	$R_{\min,M}/2$ (Å)	ϵ_M (kcal/mol)	$R_{\min,M}/2$ (Å)	ϵ_M (kcal/mol)	$R_{\min,M}/2$ (Å)	ϵ_M (kcal/mol)	$R_{\min,M}/2$ (Å)	ϵ_M (kcal/mol)
Al ³⁺	1.005	0.00001631	0.924	0.00000142	0.972	0.00000639	0.916	0.00000108
Fe ³⁺	1.121	0.00025449	1.023	0.00002635	1.099	0.00016028	1.013	0.00002024
Cr ³⁺	1.217	0.00146124	1.145	0.00040986	1.197	0.00104974	1.140	0.00037198
In ³⁺	1.241	0.00212802	1.157	0.00051476	1.211	0.00132548	1.148	0.00043416
Tl ³⁺	1.253	0.00254709	1.157	0.00051476	1.211	0.00132548	1.152	0.00046851
Y ³⁺	1.458	0.02735051	1.399	0.01554690	1.431	0.02133669	1.391	0.01430674
La ³⁺	1.646	0.10417397	1.577	0.06841702	1.624	0.09180886	1.574	0.06706518
Ce ³⁺	1.610	0.08440707	1.540	0.05292325	1.589	0.07399405	1.537	0.05177853
Pr ³⁺	1.583	0.07117158	1.509	0.04188268	1.561	0.06140287	1.506	0.04090549
Nd ³⁺	1.562	0.06182717	1.489	0.03566355	1.540	0.05292325	1.485	0.03450196
Sm ³⁺	1.531	0.04953859	1.458	0.02735051	1.512	0.04287573	1.459	0.02759452
Eu ³⁺	1.514	0.04354662	1.438	0.02279185	1.492	0.03655251	1.430	0.02113456
Gd ³⁺	1.505	0.04058327	1.431	0.02133669	1.485	0.03450196	1.421	0.01937874
Tb ³⁺	1.479	0.03280986	1.420	0.01919059	1.459	0.02759452	1.409	0.01721000
Dy ³⁺	1.467	0.02960343	1.409	0.01721000	1.446	0.02454281	1.401	0.01586934
Er ³⁺	1.433	0.02174524	1.380	0.01272679	1.415	0.01827024	1.373	0.01179373
Tm ³⁺	1.423	0.01975917	1.372	0.01166488	1.412	0.01773416	1.364	0.01067299
Lu ³⁺	1.423	0.01975917	1.372	0.01166488	1.412	0.01773416	1.364	0.01067299
Hf ⁴⁺	1.134	0.00033060	1.023	0.00002635	1.109	0.00019838	1.014	0.00002079
Zr ⁴⁺	1.169	0.00064221	1.092	0.00013761	1.150	0.00045105	1.087	0.00012321
Ce ⁴⁺	1.370	0.01141046	1.306	0.00530214	1.352	0.00930991	1.301	0.00496778
U ⁴⁺	1.247	0.00232970	1.155	0.00049581	1.223	0.00160860	1.148	0.00043416
Pu ⁴⁺	1.308	0.00544088	1.221	0.00155814	1.285	0.00401101	1.215	0.00141473
Th ⁴⁺	1.475	0.03171494	1.393	0.01460944	1.454	0.02639002	1.388	0.01386171

Table 3. 12-6 IOD Parameter Set for 18 Trivalent and 6 Tetravalent Metal Ions in Conjunction with OPC3, OPC, TIP3P-FB, and TIP4P-FB Water Models

	OPC3		OPC		TIP3P-FB		TIP4P-FB	
	$R_{\min,M}/2$ (Å)	ϵ_M (kcal/mol)	$R_{\min,M}/2$ (Å)	ϵ_M (kcal/mol)	$R_{\min,M}/2$ (Å)	ϵ_M (kcal/mol)	$R_{\min,M}/2$ (Å)	ϵ_M (kcal/mol)
Al ³⁺	1.287	0.00412163	1.250	0.00243637	1.287	0.00412163	1.267	0.00312065
Fe ³⁺	1.419	0.01900380	1.400	0.01570749	1.419	0.01900380	1.404	0.01636246
Cr ³⁺	1.364	0.01067299	1.336	0.00770969	1.364	0.01067299	1.345	0.00858042
In ³⁺	1.453	0.02615377	1.440	0.02322071	1.445	0.02431873	1.443	0.02387506
Tl ³⁺	1.496	0.03776169	1.483	0.03393126	1.496	0.03776169	1.488	0.03537062
Y ³⁺	1.600	0.07934493	1.575	0.06751391	1.600	0.07934493	1.583	0.07117158
La ³⁺	1.733	0.16134811	1.700	0.13818331	1.731	0.15989650	1.715	0.14850170
Ce ³⁺	1.758	0.17997960	1.725	0.15557763	1.754	0.17693975	1.738	0.16500296
Pr ³⁺	1.750	0.17392181	1.717	0.14990448	1.746	0.17092614	1.731	0.15989650
Nd ³⁺	1.692	0.13282966	1.662	0.11371963	1.692	0.13282966	1.675	0.12180998
Sm ³⁺	1.667	0.11679623	1.638	0.09957472	1.667	0.11679623	1.650	0.10651723
Eu ³⁺	1.675	0.12180998	1.646	0.10417397	1.675	0.12180998	1.658	0.11129023
Gd ³⁺	1.625	0.09235154	1.600	0.07934493	1.625	0.09235154	1.608	0.08337961
Tb ³⁺	1.633	0.09675968	1.608	0.08337961	1.633	0.09675968	1.617	0.08806221
Dy ³⁺	1.608	0.08337961	1.583	0.07117158	1.608	0.08337961	1.592	0.07543075
Er ³⁺	1.600	0.07934493	1.575	0.06751391	1.600	0.07934493	1.583	0.07117158
Tm ³⁺	1.600	0.07934493	1.575	0.06751391	1.600	0.07934493	1.583	0.07117158
Lu ³⁺	1.583	0.07117158	1.558	0.06014121	1.583	0.07117158	1.567	0.06397679
Hf ⁴⁺	1.489	0.03566355	1.461	0.02808726	1.489	0.03566355	1.467	0.02960343
Zr ⁴⁺	1.508	0.04155519	1.488	0.03537062	1.508	0.04155519	1.495	0.03745682
Ce ⁴⁺	1.708	0.14364160	1.683	0.12693448	1.705	0.14158262	1.692	0.13282966
U ⁴⁺	1.708	0.14364160	1.683	0.12693448	1.705	0.14158262	1.692	0.13282966
Pu ⁴⁺	1.682	0.12628793	1.658	0.11129023	1.682	0.12628793	1.667	0.11679623
Th ⁴⁺	1.721	0.15272873	1.704	0.14089951	1.718	0.15060822	1.709	0.14433113

where the IOD parameter set always has larger $R_{\min}/2$ and ϵ values than the HFE parameter set for the same metal ion. No 12-6 compromise (CM) set was developed for the trivalent and

tetravalent metal ions, which is consistent with our previous study,⁵⁴ because of the relatively large errors for both HFE and IOD for a CM parameter set. The 12-6-4 model, as we show

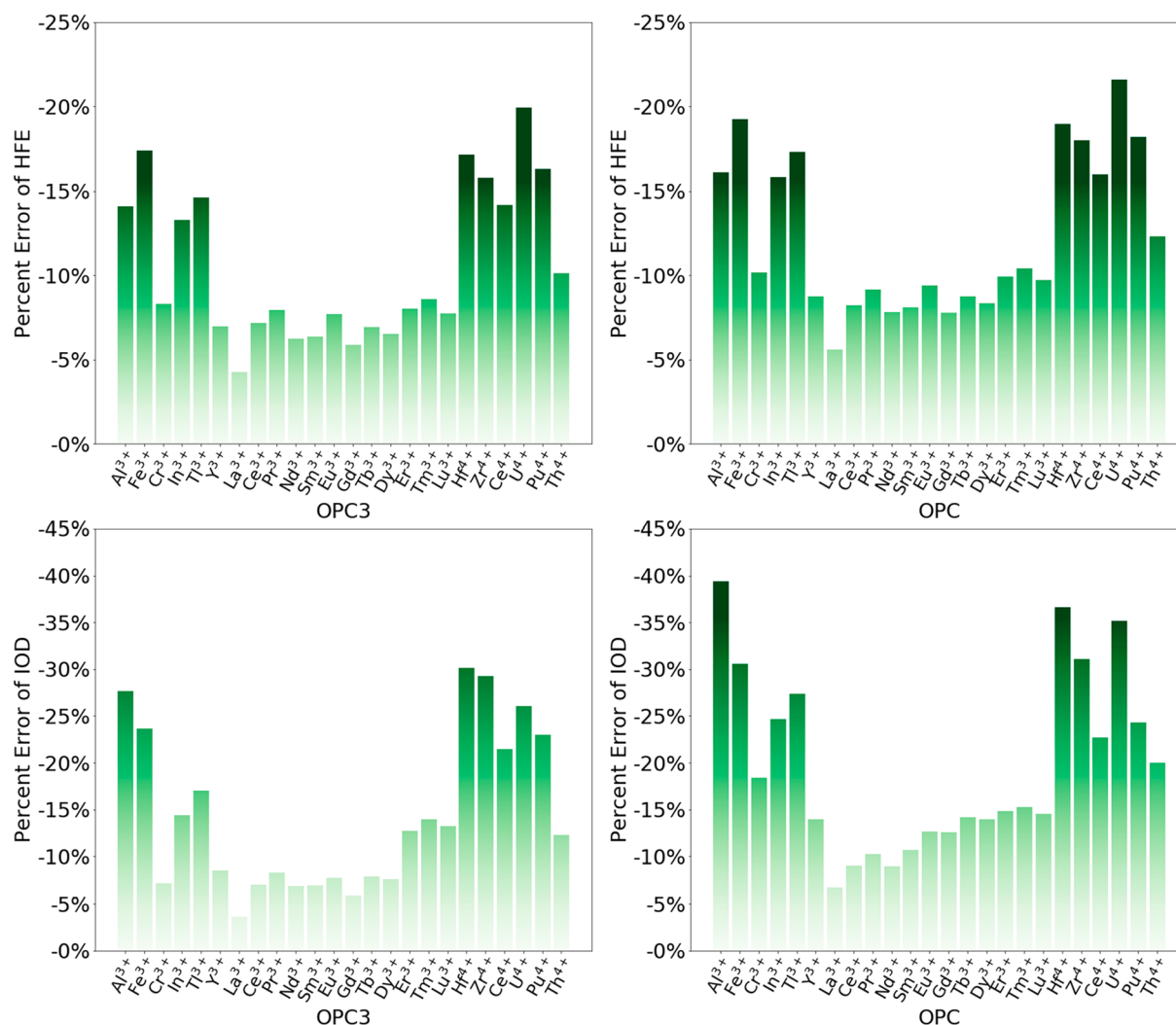


Figure 5. HFE and IOD percent errors for the 12-6 nonbonded model of highly charged metal ions in conjunction with OPC3 and OPC water models. The HFE percent errors (top two figures) were for the 12-6 IOD parameter set which can reproduce the experimental IOD values (Table 3 and Table S5). The IOD percent errors were for the 12-6 HFE parameter set which can reproduce the experimental HFE values (Table 2 and Table S4).

below, offers a much better representation of these ions in aqueous solution.

Error Analysis. Further error analysis was conducted by analyzing the simulated HFEs using the 12-6 IOD parameter set and the simulated IODs using the 12-6 HFE parameter set. The relative percentage errors of the simulated HFE and IOD values with respect to the experimental values are plotted in Figure 5 and Figure S1.

By comparing the percent errors for trivalent and tetravalent metal ions, the latter have generally larger errors than the former for both the HFE and IOD. This, apart from the fact that the absolute HFE and IOD values are larger for tetravalent ions, can also be explained by the stronger effect of the ion-induced dipole for the highly charged metal ions.

Parameterization of the 12-6-4 Model. Given the weakness of the 12-6 model, we subsequently parametrized the 12-6-4 model for the same 24 metal ions. Starting from the initial guesses obtained based on the method illustrated in Figure 1, we parametrized the 12-6-4 model for the 18 M(III) and 6 M(IV) metal ions. The obtained parameters are given in Table 4.

Features of Parametrization Results and Their Connection with Metal Ion Properties.

For the trivalent ions of the main group elements, Al^{3+} , In^{3+} , and Tl^{3+} , their C_4 parameters have a noticeable trend of $\text{Tl}^{3+} > \text{Al}^{3+} > \text{In}^{3+}$. A reasonable explanation is that Tl can have a monovalent oxidation state due to the 6s inert pair effect,⁶⁹ with the HFE of Tl^+ being similar to that of K^+ in the Marcus HFE set.⁶⁷ Because it can be partially reduced, Tl^{3+} is the only one among these three trivalent ions which has a positive reduced electric potential,⁷⁰ making it a reactive species which leads to a strong charge transfer effect between Tl^{3+} and coordinated water molecules. In comparison, the Al^{3+} ion is a “harder” ion compared to In^{3+} , which leads to more covalent-like interactions with the coordinated water molecules.

For the lanthanide ions, their C_4 terms are generally within the range of 131–326 kcal/mol·Å⁴ for all four water models, while for main group and d-orbital transition metals the range is 209–531 kcal/mol·Å⁴ also for all four water models. This agrees with what is represented in Figure 5 and Figure S1, where for a specific metal ion the larger the HFE or IOD error it has, the larger its C_4 value will be to compensate for that error. Among the actinide series tetravalent ions, U^{4+} is the only ion that has a

Table 4. 12-6-4 Parameter Set for 18 Trivalent and 6 Tetravalent Metal Ions in Conjunction with OPC3, OPC, TIP3P-FB, and TIP4P-FB Water Models

	OPC3			OPC			TIP3P-FB			TIP4P-FB		
	$R_{\text{min,M}}/2$ (Å)	ϵ_{M} (kcal/mol)	C_4 (kcal/mol·Å ⁴)	$R_{\text{min,M}}/2$ (Å)	ϵ_{M} (kcal/mol)	C_4 (kcal/mol·Å ⁴)	$R_{\text{min,M}}/2$ (Å)	ϵ_{M} (kcal/mol)	C_4 (kcal/mol·Å ⁴)	$R_{\text{min,M}}/2$ (Å)	ϵ_{M} (kcal/mol)	C_4 (kcal/mol·Å ⁴)
Al ³⁺	1.361	0.01031847	363	1.335	0.00761745	399	1.366	0.01091456	387	1.357	0.00986018	427
Fe ³⁺	1.455	0.02662782	429	1.461	0.02808726	531	1.452	0.02591906	446	1.451	0.02568588	502
Cr ³⁺	1.402	0.01603244	209	1.379	0.01259012	243	1.404	0.01636246	232	1.403	0.01619682	286
In ³⁺	1.487	0.03507938	330	1.485	0.03450196	413	1.488	0.03537062	343	1.484	0.03421577	403
Tl ³⁺	1.567	0.06397679	437	1.537	0.05177853	479	1.572	0.06617338	464	1.565	0.06311131	514
Y ³⁺	1.626	0.09289608	192	1.615	0.08700853	260	1.629	0.09454081	218	1.622	0.09072908	268
La ³⁺	1.751	0.17467422	131	1.719	0.15131351	165	1.757	0.17921760	155	1.751	0.17467422	211
Ce ³⁺	1.781	0.19786584	215	1.775	0.19313505	289	1.791	0.20584696	251	1.781	0.19786584	294
Pr ³⁺	1.785	0.20104406	255	1.765	0.18535099	311	1.791	0.20584696	291	1.777	0.19470705	326
Nd ³⁺	1.712	0.14640930	184	1.699	0.13750834	243	1.716	0.14920231	211	1.707	0.14295367	256
Sm ³⁺	1.690	0.13150785	188	1.666	0.111617738	236	1.695	0.13482489	218	1.683	0.12693448	257
Eu ³⁺	1.704	0.14089951	233	1.680	0.12499993	279	1.708	0.14364160	261	1.699	0.13750834	302
Gd ³⁺	1.644	0.10301322	164	1.627	0.09344247	222	1.648	0.10534198	191	1.639	0.10014323	238
Tb ³⁺	1.657	0.11068733	199	1.639	0.10014323	256	1.661	0.11310961	226	1.651	0.10710756	270
Dy ³⁺	1.632	0.09620220	183	1.616	0.08753443	243	1.643	0.10243558	219	1.627	0.09344247	253
Er ³⁺	1.630	0.09509276	228	1.620	0.08965674	298	1.634	0.09731901	256	1.626	0.09289608	304
Tm ³⁺	1.633	0.09675968	246	1.622	0.09072908	314	1.645	0.10359269	285	1.628	0.09399072	320
Lu ³⁺	1.616	0.08753443	222	1.605	0.08185250	289	1.619	0.08912336	250	1.615	0.08700853	303
Hf ⁴⁺	1.563	0.06225334	718	1.556	0.05930945	847	1.580	0.06978581	787	1.561	0.06140287	837
Zr ⁴⁺	1.597	0.07786298	707	1.576	0.06796452	804	1.611	0.08492362	769	1.603	0.08084383	845
Ce ⁴⁺	1.752	0.17542802	653	1.747	0.17167295	789	1.760	0.18150763	701	1.750	0.17392181	771
U ⁴⁺	1.770	0.18922704	980	1.767	0.18689752	1123	1.788	0.20344023	1044	1.787	0.20264033	1140
Pu ⁴⁺	1.755	0.17769767	817	1.745	0.17018074	941	1.744	0.16943676	817	1.744	0.16943676	919
Th ⁴⁺	1.755	0.17769767	452	1.755	0.17769767	598	1.763	0.18380968	507	1.761	0.18227365	601

Table 5. Performance of Parameters Sets Developed in the Present Study in Simulating the Diffusivity Coefficient of Fe^{3+} When Used in Conjunction with the OPC Water Model

Parameter Set	Diffusion coefficient ($10^{-5} \text{ cm}^2/\text{s}$)
12-6 HFE	0.605 ± 0.093
12-6 IOD	0.765 ± 0.106
12-6-4	0.708 ± 0.089
Experiment	0.604

C_4 about or above $1000 \text{ kcal/mol} \cdot \text{\AA}^4$, indicating that the U^{4+} ion is the ion with the strongest dipole-inducing ability among all the 24 metal ions investigated. Comparing to the experimental CNs, we found that the 12-6-4 model tends to overcoordinate the Pu^{4+} and Ce^{4+} ions, with Pu^{4+} being slightly worse than Ce^{4+} (see Table 1 and Table S6). This is probably a combined effect of inaccurate water interactions within the first solvation shell and the non-noble gas configuration of Pu^{4+} . According to Tables S4, S5, and S6, TIP3P-FB usually has the highest CN values compared to the other water models, which we hypothesize might be due to its small Lennard-Jones C_{12} term ($6.6 \times 10^5 \text{ kcal/mol} \cdot \text{\AA}^{12}$).⁶⁰

Calculation of Diffusion Coefficients of Several Trivalent Metal Ions. As a further test, the diffusion coefficient of Fe^{3+} was calculated for the OPC water model using the three parameter sets developed for Fe^{3+} . Using linear extrapolation, we obtained the diffusion coefficient at infinite dilution (which corresponds to the extrapolation intercept), with the results illustrated in Figure 6 and Table 5. In comparison, the experimental diffusion coefficient for Fe^{3+} is $6.04 \times 10^{-10} \text{ m}^2/$

s (i.e., $0.64 \times 10^{-5} \text{ cm}^2/\text{s}$).⁷¹ The diffusion coefficients for Cr^{3+} and Al^{3+} in conjunction with the OPC water model were also calculated and shown in Figure S4 and Table S7.

In general, all three parameter sets gave reasonable diffusion coefficients. Among the three parameter sets, we find that the 12-6 HFE parameter set yields the closest values to the experiment, while 12-6-4 was higher, and IOD was even higher. Interestingly, unlike Fe^{3+} and Cr^{3+} , the diffusion coefficient of Al^{3+} was overestimated by $\sim 10\%$ by the 12-6 HFE parameter set, which we hypothesize is due to the tendency of Al^{3+} to form $\text{Al}_m(\text{OH})_n^{(3m-n)+}$ complexes that reduces the diffusion rate under experimental conditions. In general, we find that the 12-6-4 model can reasonably reproduce experimental HFE, IOD, and diffusion coefficient simultaneously.

Application to an Archaea Superoxide Reductase. To further test our parameters, we performed MD simulations on superoxide reductase, which is an iron-containing metalloprotein system. The crystal structure of superoxide reductase from *Nanoarchaeum equitans* was obtained from the protein data bank (PDB ID, 4BV1; resolution, 1.90 Å). Superoxide reductase is a homotetramer, and each monomer has an Fe^{3+} ion binding site at the surface (Figure 7). Each of the binding sites of chains A, B, and D consists of an Fe^{3+} ion, four His residues, and one Cys residue, while the binding site of chain C has an additional water coordinating with the Fe^{3+} ion. In total, we performed nine independent MD simulations on this protein using different ion parameters combined with different water models, i.e., the 12-6-4, 12-6 HFE, and 12-6 IOD parameter sets for each of the OPC3, OPC, and TIP3P water models. Each simulation consists of 100 ns sampling, and details of the simulation procedure can be

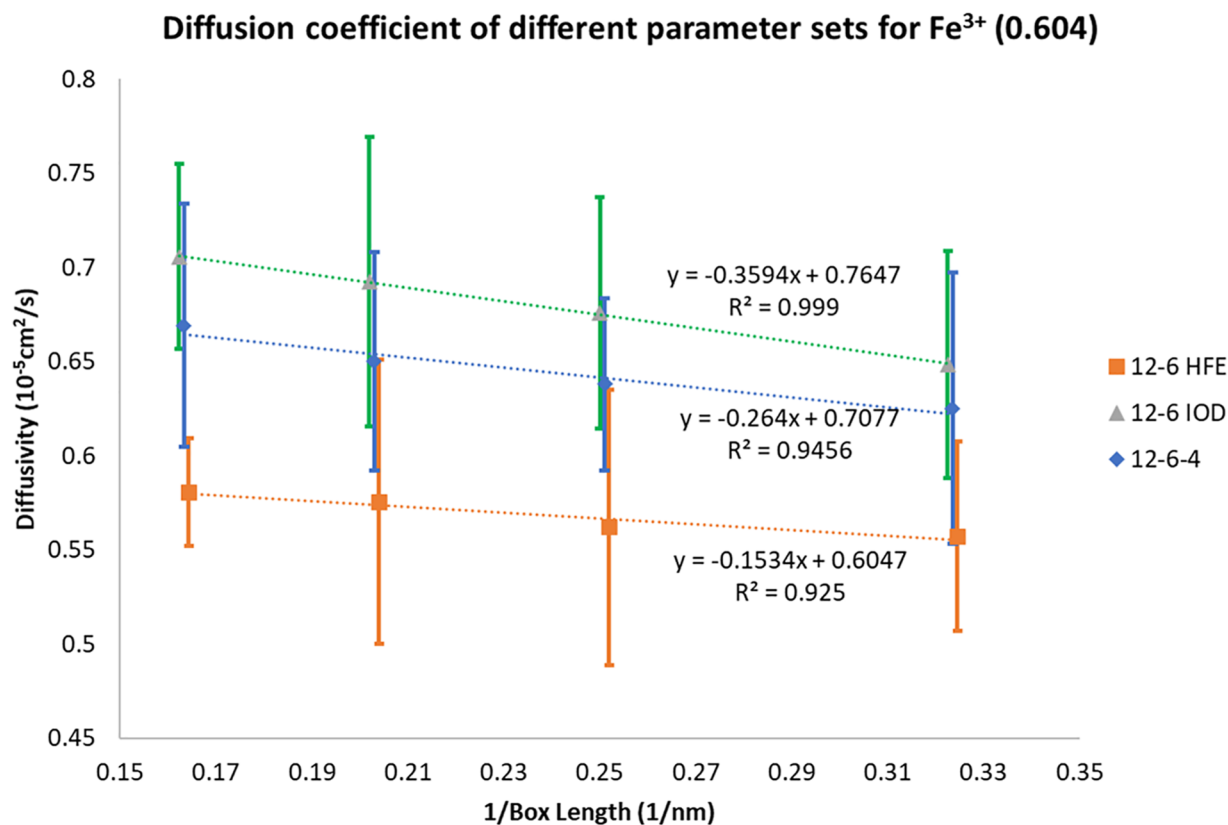


Figure 6. Diffusion coefficient of Fe^{3+} under different ion concentrations. Error bar is the standard deviation over multiple simulations, while the dashed lines are linear extrapolations for each parameter set, with their Y-intercepts being the final diffusion coefficients.

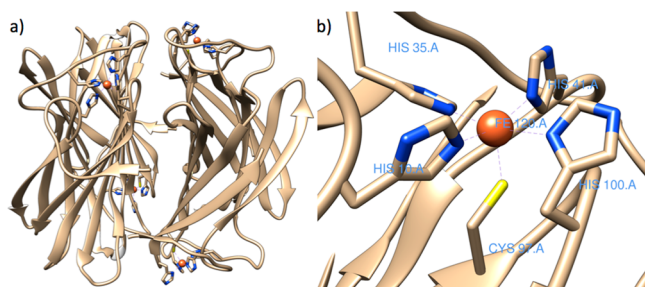


Figure 7. (a) Overall structure of superoxide reductase and (b) metal ion binding site in chain A, with metal ion and coordinating residues labeled. This figure was created based on the crystal structure by using Chimera.

found in the SI. For the 12-6-4 parameters, the C_4 terms between the metal ion and all the other atom types except the water oxygen are calculated through eq 8, where $C_4(\text{H}_2\text{O})$, $\alpha_0(\text{H}_2\text{O})$, and $\alpha_0(\text{atom type})$ are the C_4 term between the metal ion and water oxygen, the polarizability of the water oxygen (which was treated as 1.444 Å), and the polarizability of the corresponding atom type, respectively. Here, we used the default polarizabilities from the AMBER software package.⁷²

$$C_4(\text{atom type}) = \frac{C_4(\text{H}_2\text{O})}{\alpha_0(\text{H}_2\text{O})} \times \alpha_0(\text{atom type}) \quad (8)$$

RMSD analyses on the protein backbone heavy atoms and on the binding site residues (the Fe^{3+} ion, four His residues, and one Cys residue) were performed to assess the performance of the metal ion parameters (Figures 8 and 9 and Figures S3 and S4). From the RMSD plots, we can see that except for the HFE/

OPC, HFE/OPC3, and IOD/TIP3P combinations, the RMSDs of the protein backbone heavy atoms fluctuated around 0.8 Å, and the RMSDs of the binding site residues fluctuated around 0.6 Å. The small RMSDs agreed well with the retaining of the binding site structures over the course of the simulations, which were validated by visualizing the trajectories. All the Fe^{3+} ions were found to remain bound with the His and Cys residues, although a water molecule also bound with the ion in each site forming an octahedral-like structure. While for the above-mentioned HFE/OPC, HFE/OPC3, and IOD/TIP3P combinations, multiple water molecules replaced the His residues during the 100 ns simulations, although the Cys residues stay bound with the Fe^{3+} ions. It is interesting to note that for TIP3P it was the IOD parameter set which did not retain the binding site, while for the new water models (OPC and OPC3) it was the HFE set. These observations suggest that for metal ion binding in proteins, the thermodynamics are complicated and are delicately balanced among ion–water, ion–protein, protein–water, protein–protein, and water–water interactions.⁵³ In comparison, the 12-6-4 parameters perform consistently well across the different water models in terms of retaining the active site structures, which could be a better choice for simulating metalloprotein systems than the 12-6 model.

CONCLUSIONS

In the present study, we parametrized 18 trivalent and 6 tetravalent ions in conjunction with four new water models (OPC3,⁵⁸ OPC,⁵⁹ TIP3P-FB,⁶⁰ and TIP4P-FB⁶⁰). To meet different needs, we separately parametrized the 12-6 HFE, 12-6 IOD, and 12-6-4 parameter sets, which aimed to reproduce the experimental HFEs, IODs, and both of them, respectively.

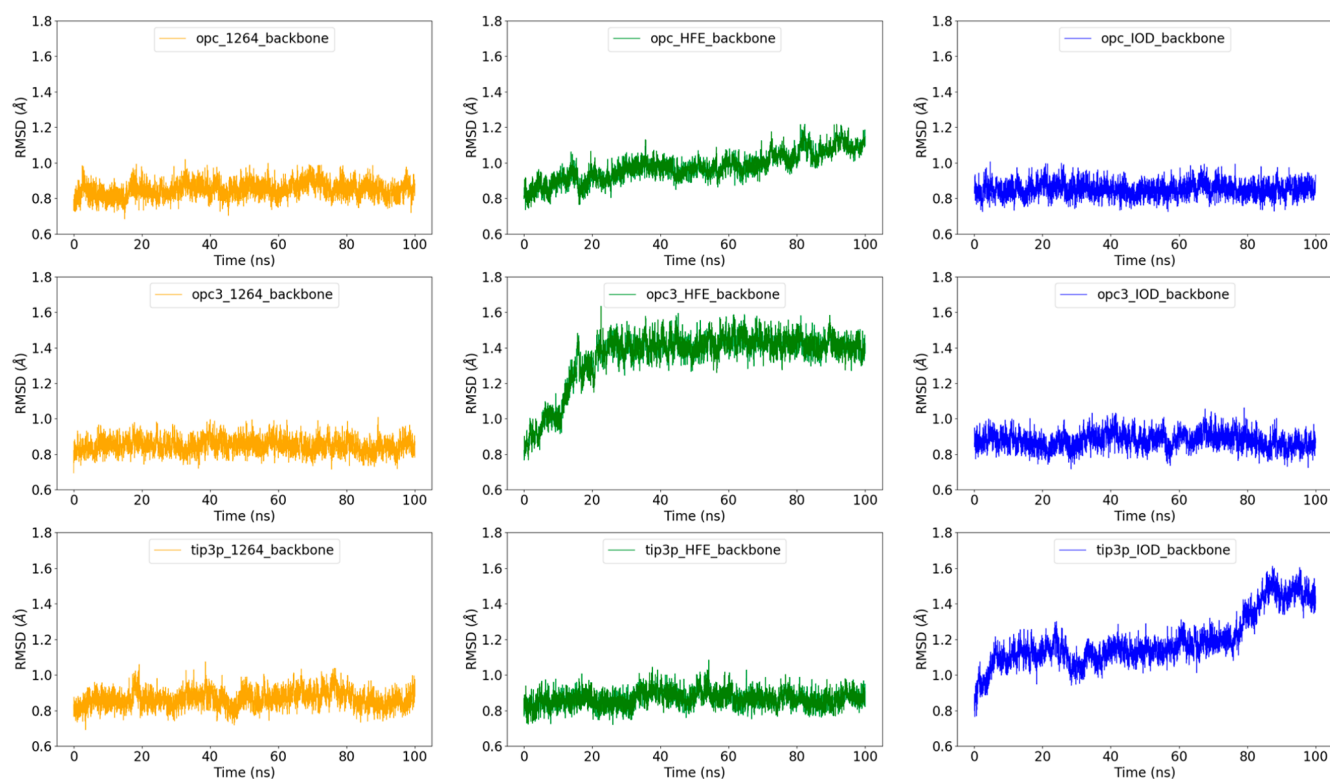


Figure 8. RMSDs of protein backbone heavy atoms for nine independent simulations which used different ion parameters and water models. The RMSDs were calculated against the initial coordinates generated based on the crystal structure (see SI).

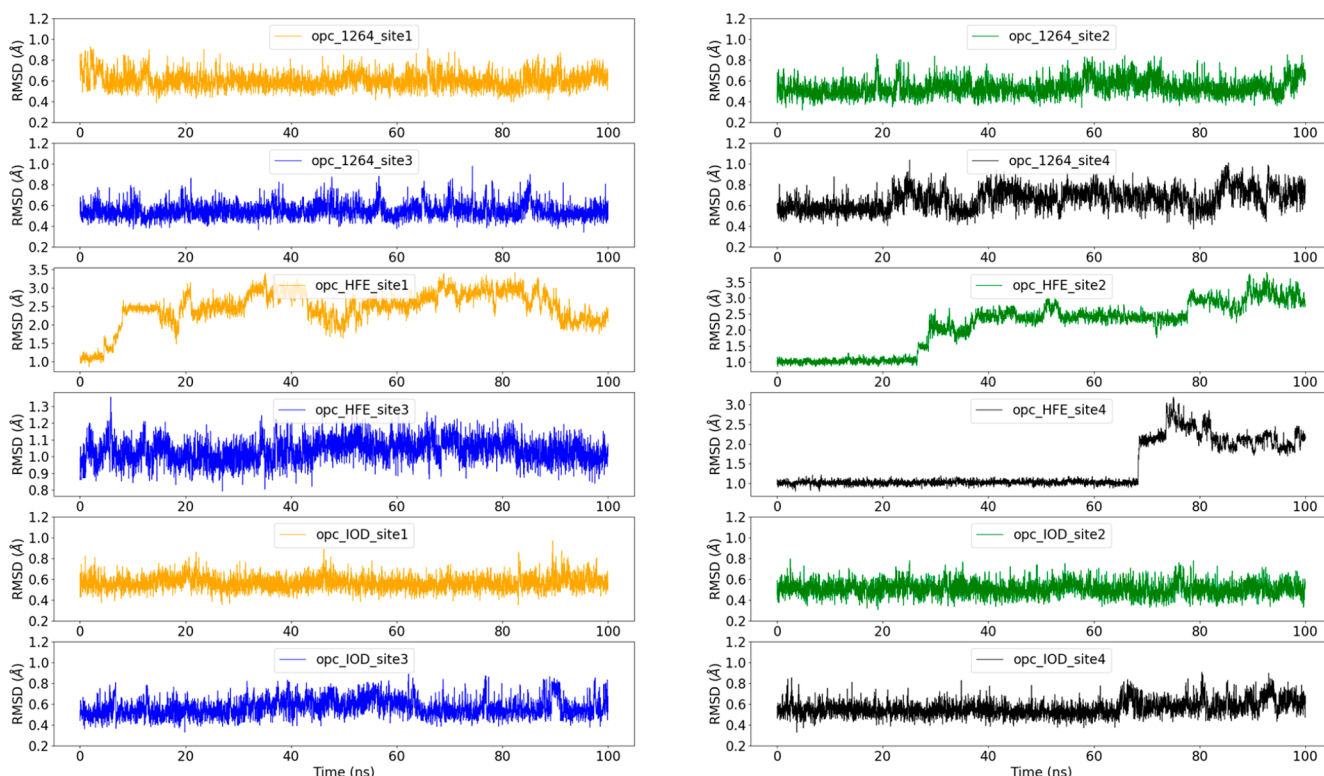


Figure 9. RMSDs of binding site residues from simulations using 12-6-4, 12-6 HFE, and 12-6 IOD parameter sets for Fe^{3+} in conjunction with the OPC water model. The RMSDs were calculated against the initial coordinates generated based on the crystal structure (see SI).

We examined the performance of the 12-6 nonbonded model to simulate highly charged metal ions in conjunction with the four new water models as well as three previous and commonly used water models (TIP3P, SPC/E, and TIP4P_{EW}). Our results indicated that these four new water models provide comparable or significantly better performance than the previous water models in the same category (between 3-point and 4-point water models, respectively). This agrees with the notion that these new water models represent improvements over earlier water models.⁶¹ In general, the three-point water models yield better results than the four-point water models when using the 12-6 model, with the OPC3 water model providing the best performance for simulating trivalent and tetravalent ions in aqueous solution. The above observations agree with our previous study of divalent metal ions.⁵¹

In the present study, comparing to the 12-6 HFE and 12-6 IOD parameter sets, the 12-6-4 model showed significant improvement with being able to simultaneously reproduce experimental HFEs within 2 kcal/mol and experimental IODs within 0.01 Å for the trivalent and tetravalent metal ions. Moreover, we also carried out simulations to evaluate the diffusion coefficients of Fe^{3+} , Al^{3+} , and Cr^{3+} . All of the 12-6 HFE, IOD, and 12-6-4 parameter sets gave a reasonable agreement with the experimental value, indicating that our parameters are well designed and are transferable. Finally, we simulated the superoxide reductase from *Nanoarchaeum equitans* based on different water models and parameter sets. Overall, most of the simulations were able to represent the metal site coordination environment with the 12-6-4 model showing better consistency in retaining the coordination sphere. These results further indicate that the 12-6-4 model is superior to the 12-6 model for

simulating metal ions. We hope our parameters will facilitate the simulation of systems containing highly charged metal ions.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jctc.0c01320>.

Methods for simulating the archaea superoxide reductase; supporting figures and tables (PDF)

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Notes

The authors declare no competing financial interest.

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